

UNIT 4: Structured populations

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Introduction

- ▶ Up until now we've tracked populations with a single state variable (population size or population density)
- ▶ What assumption are we making?
 - ▶ * All individuals are similar enough to be counted as if they are the same
 - ▶ * Always (continuous time)
 - ▶ * At census time (discrete time)
- ▶ What are some organisms for which this seems like a good approximation?
 - ▶ * Short-lived and/or don't grow much
 - ▶ * Dandelions, bacteria, insects
- ▶ What are some organisms that don't work so well?
 - ▶ * Trees, people, codfish

Structured populations

- ▶ If we think age or size is important to understanding a population, we might model it as an **structured** population
- ▶ Instead of just keeping track of the total number of individuals in our population ...
 - ▶ Keeping track of how many individuals of each age
 - ▶ or size
 - ▶ or developmental stage

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Example: biennial dandelions

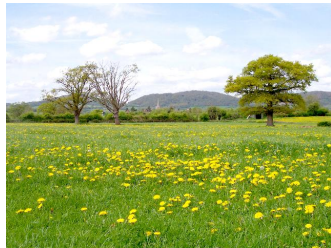
- ▶ Imagine a population of dandelions
 - ▶ Adults produce 80 seeds each year
 - ▶ 1% of seeds survive to become adults
 - ▶ 50% of first-year adults survive to reproduce again
 - ▶ Second-year adults never survive
- ▶ Will this population increase or decrease through time?

How to study this population

- ▶ Choose a census time
 - ▶ Before reproduction or after
 - ▶ Since we have complete cycle information, either one should work
- ▶ Figure out how to predict the population at the next census

Census choices

- ▶ Before reproduction
 - ▶ All individuals are adults
 - ▶ We want to know how many adults we will see next year
- ▶ After reproduction
 - ▶ Seeds, one-year-olds and two-year-olds
 - ▶ Two-year-olds have already produced their seeds; once these seeds are counted, the two-year-olds can be ignored, since they will not reproduce or survive again



Example: biennial dandelions (repeat)

- ▶ Imagine a population of dandelions
 - ▶ Adults produce 80 seeds each year
 - ▶ 1% of seeds survive to become adults
 - ▶ 50% of first-year adults survive to reproduce again
 - ▶ Second-year adults never survive
- ▶ Will this population increase or decrease through time?

What determines λ ?

- ▶ If we have 20 adults *before* reproduction, how many do we expect to see next time?
- ▶ $\lambda = p + f$ is the total number of individuals per individual after one time step
- ▶ What is f in this example?
 - ▶ * 0.8
- ▶ What is p in this example?
 - ▶ * 0.5 for 1-year-olds and 0 for 2-year-olds.
 - ▶ * It can change from year to year!

Long-term growth

- ▶ λ changes when the population structure changes
- ▶ Define $\bar{\lambda}$ as the long-term average rate of growth.
 - ▶ $N \approx N_0 \bar{\lambda}^T$
- ▶ Can we calculate $\bar{\lambda}$ if p is changing?
 - ▶ * We can't take an average over ages until we know the population structure
 - ▶ * Not easy to find the population structure until we know $\bar{\lambda} \dots$

What determines $\mathcal{R}$?

- ▶ $\mathcal{R}$ is the average total number of offspring produced by an individual over their lifespan
- ▶ Can start at any stage, but need to close the loop
- ▶ What is the reproductive number?
- ▶ * If you become an adult you produce (on average)
 - ▶ *Blackboard!*
 - ▶ * 0.8 adults your first year
 - ▶ * 0.4 adults your second year
- ▶ * $\mathcal{R} = 1.2$

What does $\mathcal{R}$ tell us about $\bar{\lambda}$?

- ▶ $\mathcal{R}$ is easier to calculate than $\bar{\lambda}$
- ▶ What is the connection?
 - ▶ * Population increases when $\mathcal{R} > 1$, so $\bar{\lambda} > 1$ exactly when $\mathcal{R} > 1$
- ▶ If $\mathcal{R} = 1.2$, then $\bar{\lambda}$
 - ▶ * > 1 – the population is increasing
 - ▶ * < 1.2 – the life cycle takes more than 1 time step, so it takes more than one time step for the population to increase 1.2 times

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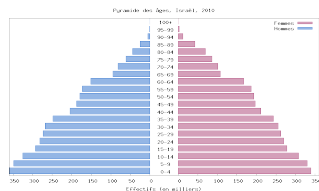
Other structured models

Stage structure

Regulated growth

Modeling approach

- ▶ In this unit, we will construct *simple* models of structured populations
 - ▶ To explore how structure might affect population dynamics
 - ▶ To investigate how to interpret structured data



Regulation

- ▶ *Simple* population models with regulation can have extremely complicated dynamics
- ▶ *Structured* population models with regulation can have insanely complicated dynamics
- ▶ Here we will focus on understanding structured population models *without regulation*:
 - ▶ * Individuals behave independently, meaning. . .
 - ▶ * Average per capita rates do not depend on population size

Age-structured models

- ▶ The most common approach is to structure by age
- ▶ In age-structured models we model how many individuals there are in each “age class”
 - ▶ Typically, we use age classes of one year
 - ▶ Example: salmon live in the ocean for roughly a fixed number of years; if we know how old a salmon is, that strongly affects how likely it is to reproduce



Stage-structured models

- ▶ In stage-structured models, we model how many individuals there are in different stages
 - ▶ i.e., newborns, juveniles, adults
 - ▶ More flexible than an age-structured model
 - ▶ Example: forest trees may survive on very little light for a long time before they have the opportunity to recruit to the sapling stage



Discrete vs. continuous time in unstructured models

- ▶ continuous-time models are structurally simpler (and smoother)
- ▶ discrete-time models only need to assume everyone's the same sometimes
 - ▶ * At the census time
 - ▶ * More realistic

...in structured models

- ▶ We no longer assume everyone is the same (we keep track of age or size)
- ▶ So it should be mostly about reproduction
 - ▶ * Populations with continuous reproduction (e.g. bacteria), may be better suited to continuous-time models
 - ▶ * Populations with **synchronous** reproduction (e.g., moths) may be better suited to discrete-time models
- ▶ Continuous time with structure gives people headaches
 - ▶ So we won't do it here, only for the sake of simplicity (and time)

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Constructing a model

- ▶ This section will focus on **linear, discrete-time, age-structured** models
- ▶ State variables: how many individuals of each age at any given time
- ▶ Parameters: p and f for each age that we are modeling

When to count

- ▶ We will choose a census time that is appropriate for our study
 - ▶ Before reproduction, to have the fewest number of individuals
 - ▶ After reproduction, to have the most information about the population processes
 - ▶ Some other time, for convenience in counting
 - ▶ * A time when individuals gather together
 - ▶ * A time when they are easy to find (insect pupae)

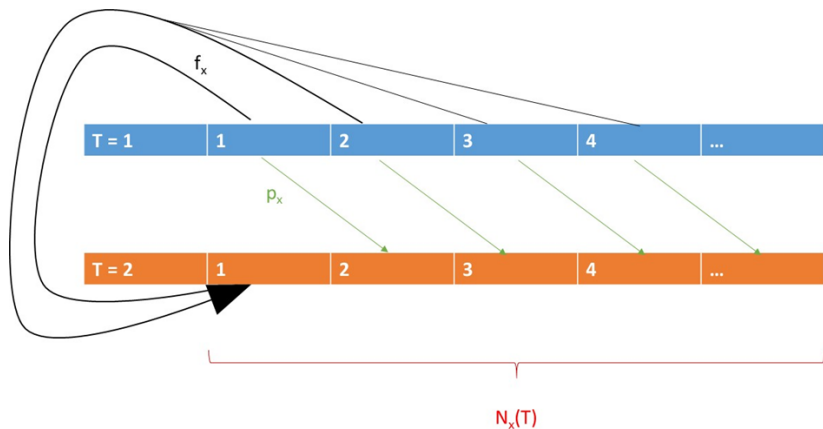
The conceptual model

- ▶ Once we choose a census time, we imagine we know the population for each age x after time step T .
 - ▶ We call these values $N_x(T)$
- ▶ Now we want to calculate the expected number of individuals in each age class at the next time step
 - ▶ We call these values $N_x(T + 1)$
- ▶ What are the parameters? — What do we need to know to calculate population for next time?
 - ▶ * The survival probability of each age group: p_x
 - ▶ * The average fecundity of each age group: f_x

Closing the loop

- ▶ f_x and p_x must close the loop back to the census time, so we can use them to simulate our model:
 - ▶ f_x has units [new indiv (at census time)]/[age x indiv (at census time)]
 - ▶ p_x has units [age $x + 1$ indiv (at census time)]/[age x indiv (at census time)]

The structured model



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Short-term dynamics

- ▶ This model's short-term dynamics will depend on parameters
 - ...
 - ▶ It is more likely to go up if fecundities and survival probabilities are high
- ▶ ...and starting conditions
 - ▶ If we start with mostly very old or very young individuals, it might go down; with lots of reproductive adults it might go up

Long-term dynamics

- ▶ If a population follows a model like this, it will tend to reach
 - ▶ a **stable age distribution**:
 - ▶ the *proportion* of individuals in each age class is constant
 - ▶ a stable value of λ_T
 - ▶ * Multiplicative growth each step approaches a constant
 - ▶ if the proportions are constant, then we can average over f_x and p_x , and the system will behave like our simple model
- ▶ What are the long-term dynamics of such a system?
 - ▶ * Exponential growth or exponential decline
- ▶ Spreadsheet link
- ▶ Add Update year

Exception

- ▶ Populations with **independent cohorts** do not tend to reach a stable age distribution
 - ▶ A **cohort** is a group that enters the population at the same time
 - ▶ We say my cohort and your cohort interact if my children might be in the same cohort as your children
 - ▶ or my grandchildren might be in the same cohort as your great-grandchildren
 - ▶ ...
- ▶ As long as all cohorts interact (none are independent), then the unregulated model leads to a stable age distribution (SAD)

Independent cohorts

- ▶ Some populations might have independent cohorts:
 - ▶ If salmon reproduce *exactly* every four years, then:
 - ▶ the 2015 cohort would have offspring in 2019, 2023, 2027, 2031, ...
 - ▶ the 2016 cohort would have offspring in 2020, 2024, 2028, 2032, ...
 - ▶ in theory, they could remain independent – distribution would not converge
- ▶ Examples could include 17-year locusts, century plants, ...

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Life tables

- ▶ People often study structured models using **life tables**
- ▶ A life table is made *from the perspective of a particular census time*
- ▶ It contains the information necessary to project to the next census:
 - ▶ How many survivors do we expect at the next census for each individual we see at this census? (p_x in our model)
 - ▶ How many offspring do we expect at the next census for each individual we see at this census? (f_x in our model)

Cumulative survivorship

- ▶ The first key to understanding how much each organism will contribute to the population is **survivorship**
- ▶ In the field, we estimate the probability of survival from age x to age $x + 1$: p_x
 - ▶ This is the probability you will be *counted* at age $x + 1$, given that you were *counted* at age x .
- ▶ To understand how individuals contribute to the population, we are also interested in the overall probability that individuals survive to age x : ℓ_x .
 - ▶ * $\ell_x = p_1 \times \dots \times p_{x-1}$
 - ▶ * ℓ_x measures the probability that an individual survives to be counted at age x , given that it is ever counted at all (ie., it survives to its first census)

Calculating $\mathcal{R}$

- ▶ We calculate $\mathcal{R}$ by figuring out the estimated contribution at each age group, *per individual who was ever counted*
 - ▶ We figure out expected contribution given you were ever counted by multiplying:
 - ▶ * $f_x \times \ell_x$

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Dandelion example



Example: biennial dandelions (repeat)

- ▶ Adults produce 80 seeds each
- ▶ 1% of seeds survive to become adults
- ▶ 50% of first-year adults survive to reproduce again
- ▶ Second-year adults never survive
- ▶ What does the life table look like?

Dandelion life table

x	f_x	p_x	l_x	$l_x f_x$
1				
2				
R				

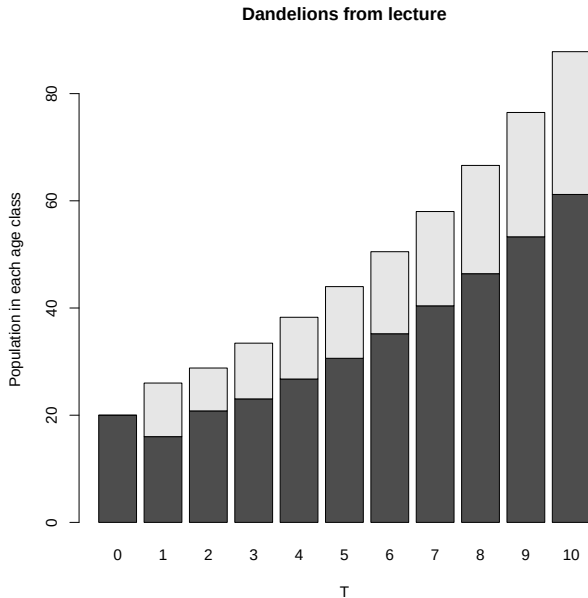
Dandelion life table (repeat)

x	f_x	p_x	l_x	$l_x f_x$
1	0.8	0.5		
2	0.8	0		
R				

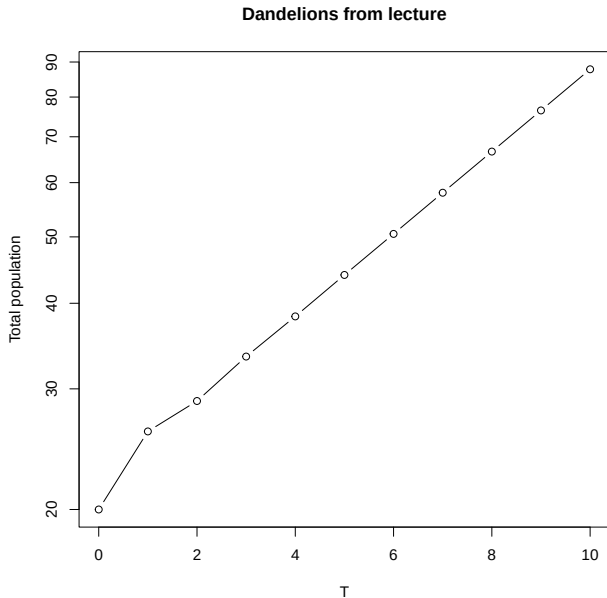
Dandelion life table

x	f_x	p_x	l_x	$l_x f_x$
1	0.8	0.5	1.000	0.800
2	0.8	0	0.500	0.400
R				1.200

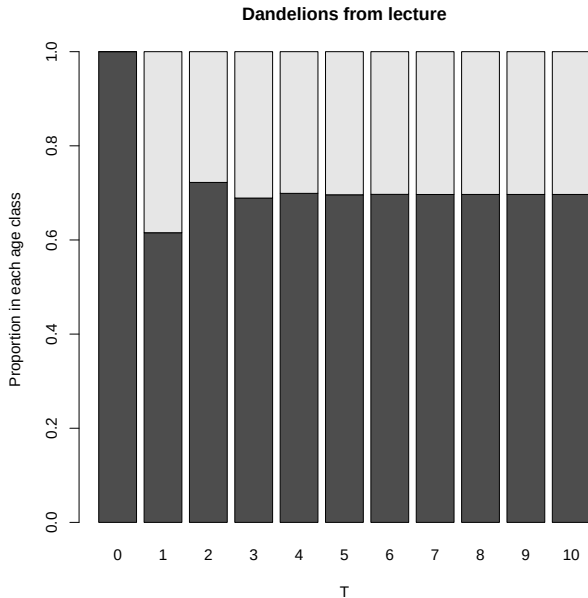
Dandelion dynamics



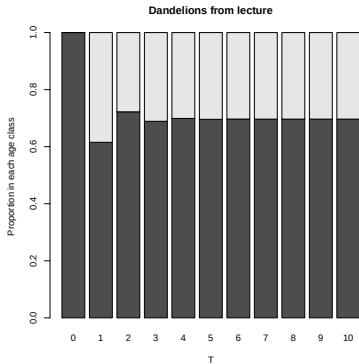
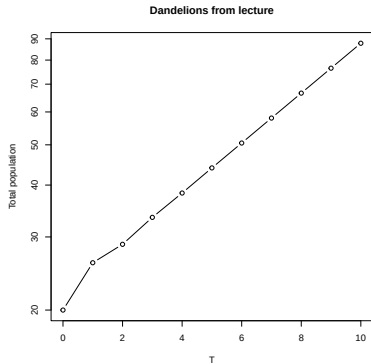
Dandelion population dynamics (present)



Dandelion age dynamics (present)



Dandelion dynamics



Squirrel example



Gray squirrel population example

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.25		
2	1.28	0.46		
3	2.28	0.77		
4	2.28	0.65		
5	2.28	0.67		
6	2.28	0.64		
7	2.28	0.88		
8	2.28	0.0		
R				

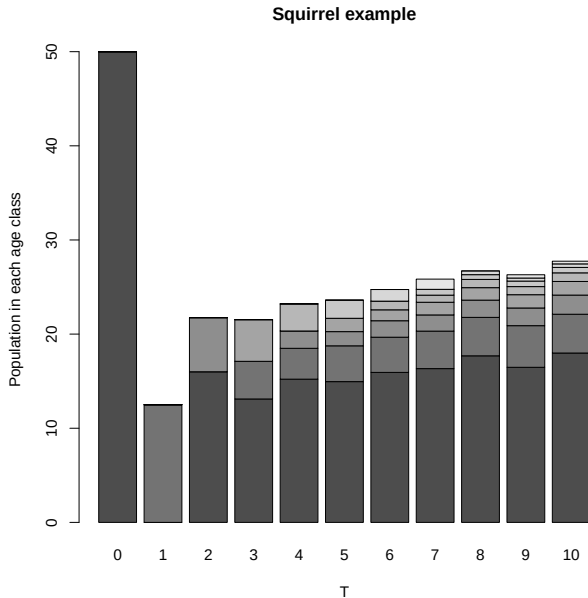
Squirrel observations

- ▶ Do you notice anything strange?
 - ▶ * Older age groups seem to be grouped for fecundity.
 - ▶ * Strange pattern in survivorship; do we really believe nobody survives past the last year?
 - ▶ * Might be better to use a model where they keep track of first year, second year, and “adult” – not much harder.
 - ▶ * This is what we call stage structure

Gray squirrel population example

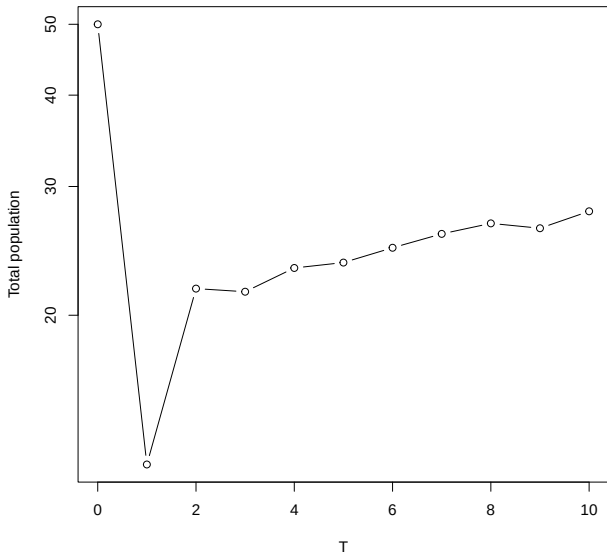
x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.25	1.000	0.000
2	1.28	0.46	0.250	0.320
3	2.28	0.77	0.115	0.262
4	2.28	0.65	0.089	0.202
5	2.28	0.67	0.058	0.131
6	2.28	0.64	0.039	0.088
7	2.28	0.88	0.025	0.056
8	2.28	0.0	0.022	0.050
R				1.109

Gray squirrel dynamics

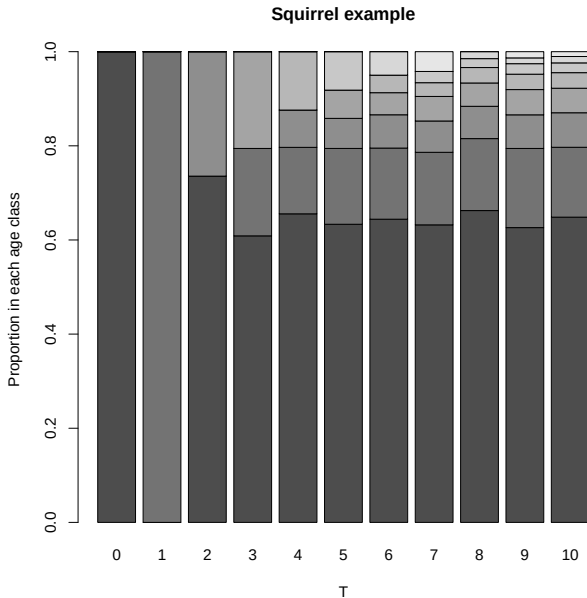


Gray squirrel population dynamics (present)

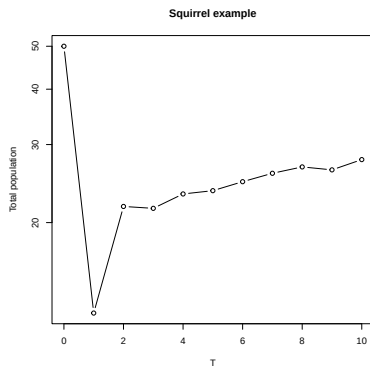
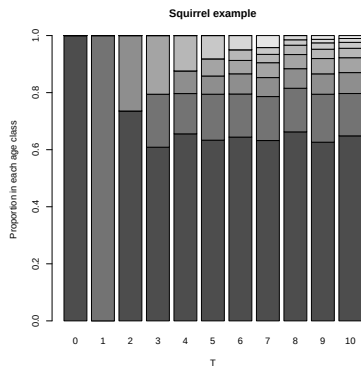
Squirrel example



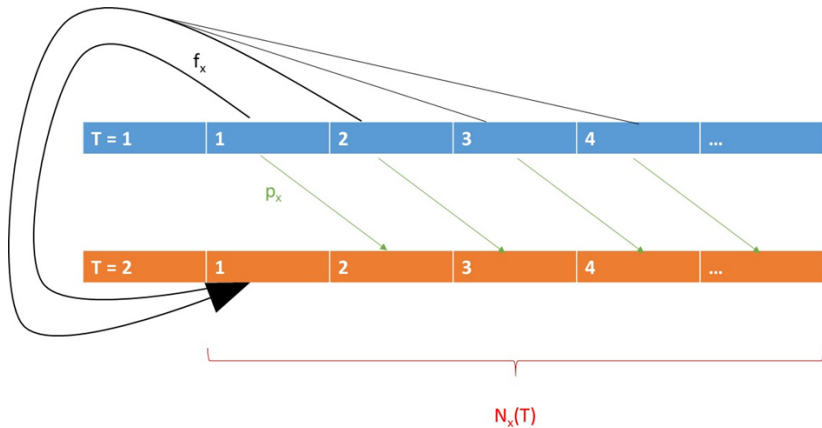
Gray squirrel age dynamics (present)



Gray squirrel dynamics



The structured model (repeat)



Salmon example

- ▶ What happens when a population has independent cohorts?
 - ▶ Does not necessarily converge to a SAD

QSLIDE Salmon example

x	f_x	p_x	l_x	$l_x f_x$
1	0	0.1		
2	0	0.5		
3	0	0.6		
4	50	0		
R				

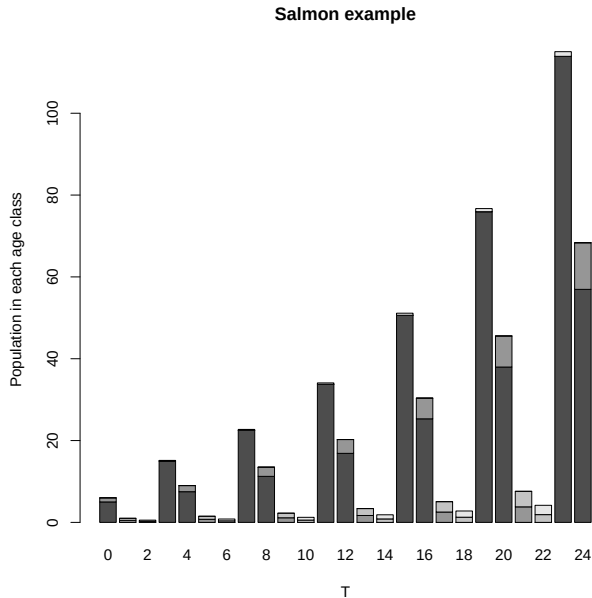
ASLIDE Salmon example

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.1	1.000	0.000
2	0	0.5	0.100	0.000
3	0	0.6	0.050	0.000
4	50	0	0.030	1.500
R				1.500

PSLIDE Salmon example

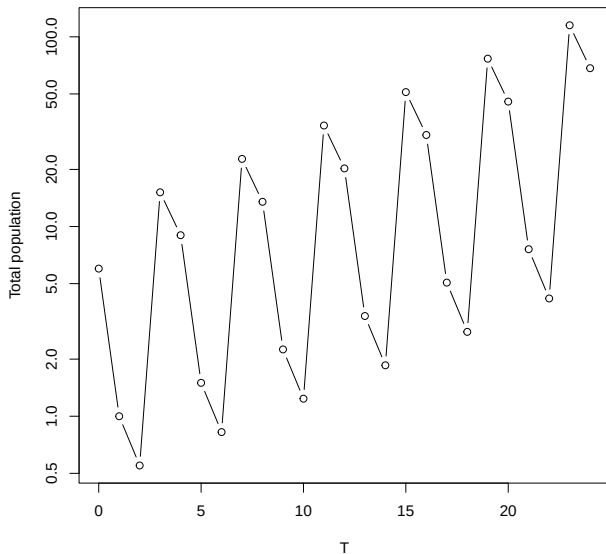


PSLIDE Salmon dynamics

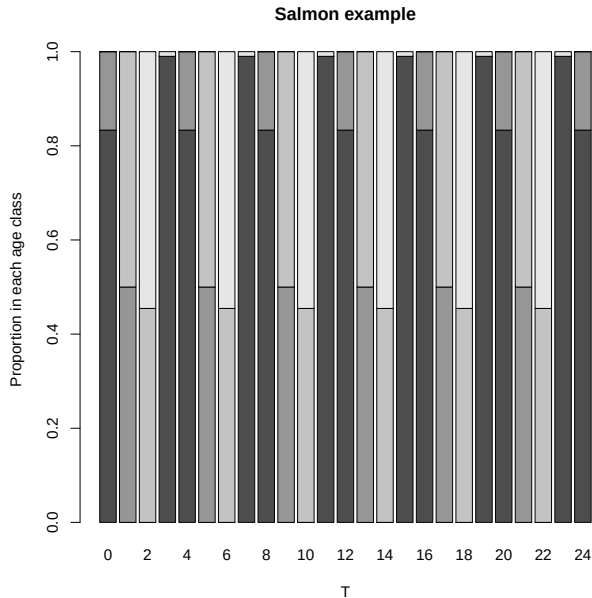


PSLIDE Salmon population dynamics

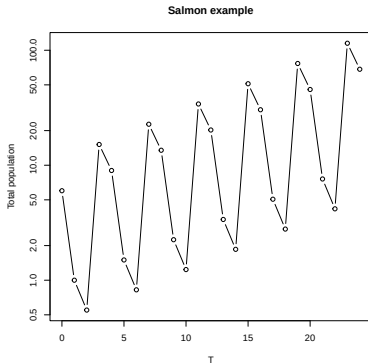
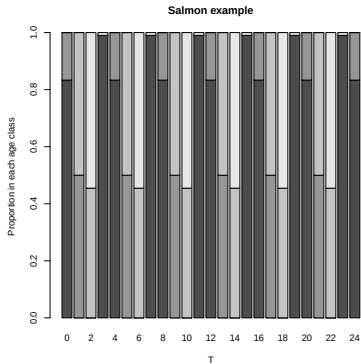
Salmon example



PSLIDE Salmon age dynamics



Salmon dynamics



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f_x vs. m_x

- ▶ Here we focus on f_x – the number of offspring seen at the next census (next year) per organism of age x seen at this census
- ▶ An alternative perspective is m_x : the total number of offspring per reproducing individual of age x
- ▶ How would I calculate one from the other?
 - ▶ * To get f_x we multiply m_x by one or more survival terms, depending on when the census is
 - ▶ * f_x needs to close the loop from one census to the next

When do we start counting?

- ▶ Is the first age class called 0, or 1?
 - ▶ In this course, we will start from age class 1
 - ▶ How many times have we seen you (not, how old are you)?
 - ▶ If we count right *after* reproduction, this means we are calling newborns age class 1.
 - ▶ If we count right *before* reproduction
 - ▶ * Age class 1 is about one year old the first time we see them

Example: biennial dandelions (repeat)

- ▶ Adults produce 80 seeds each (m_x)
- ▶ 1% of seeds survive to become adults
- ▶ 50% of first-year adults survive to reproduce again
- ▶ Second-year adults never survive
- ▶ What does the life table look like?

Dandelion life table

x	f_x	p_x	l_x	$l_x f_x$
1	0.8	0.5	1.000	0.800
2	0.8	0	0.500	0.400
R				1.200

Counting after reproduction

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0.8	0.01	1.000	0.800
2	40	0	0.010	0.400
R				1.200

There are two different approaches to the third age class: if we assume that we count the two-year old adults ($x = 3$), we can write $p_2 = 0.5$; $f_3 = 0$, and get the same answer (with one extra row that has zero contribution).

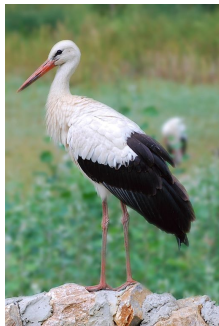
Calculating $\mathcal{R}$

- ▶ The reproductive number $\mathcal{R}$ gives the average lifetime reproduction of an individual, and is a valuable summary of the information in the life table
 - ▶ $\mathcal{R} = \sum_x \ell_x f_x$
 - ▶ If $\mathcal{R} > 1$ in the long (or medium) term, the population will increase
 - ▶ If $\mathcal{R}$ is persistently < 1 , the population is in trouble
- ▶ We can ask (for example):
 - ▶ Which ages have a large *contribution* to $\mathcal{R}$?
 - ▶ How does $\mathcal{R}$ respond to changes in various p_x and f_x ?

The effect of old individuals

- ▶ Estimating the effects of old individuals on a population can be difficult, because both f and ℓ can be extreme
 - ▶ The contribution of an age class to $\mathcal{R}$ is $\ell_x f_x$
 - ▶ Extreme how?
 - ▶ * In most populations ℓ can be very small for large x
 - ▶ * In many populations, f can be very large for large x
- ▶ Reproductive potential of old individuals *may* or *may not* be important
 - ▶ * In many tree populations, most individuals don't survive to get huge, but the huge trees may have most of the total reproduction
 - ▶ * In many bird populations, old birds produce well, but not enough to outweigh the low probability of surviving to get old.

Old individuals



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The Euler equation

- ▶ If the life table doesn't change, then $\bar{\lambda}$ is given by
$$\sum_x \ell_x f_x \bar{\lambda}^{-x} = 1$$
- ▶ We basically ask, if the population has the structure we would expect from growing at rate λ , would it continue to grow at rate λ .
- ▶ On the left-side each cohort started as λ times smaller than the one after it
 - ▶ Then got multiplied by ℓ_x .
- ▶ Under this assumption, is the next generation λ times bigger again?
- ▶ Example from spreadsheet

$\bar{\lambda}$ and $\mathcal{R}$

- ▶ If the life table doesn't change, then $\bar{\lambda}$ is given by $\sum_x \ell_x f_x \bar{\lambda}^{-x} = 1$
 - ▶ What's the relationship between $\bar{\lambda}$ and $\mathcal{R}$?
- ▶ When $\bar{\lambda} = 1$, the left hand side is just $\mathcal{R}$.
 - ▶ If $\mathcal{R} > 1$, the population more than replaces itself when $\bar{\lambda} = 1$. We must make $\bar{\lambda} > 1$ to decrease LHS and balance.
 - ▶ If $\mathcal{R} < 1$, the population fails to replace itself when $\bar{\lambda} = 1$. We must make $\bar{\lambda} < 1$ to increase LHS and balance.
- ▶ So $\mathcal{R}$ and $\bar{\lambda}$ tell the same story about whether the population is increasing

Time scales

- ▶ $\bar{\lambda}$ gives the number of individuals per individual *every year*
 - ▶ In the long term
- ▶ $\mathcal{R}$ gives the number of individuals per individual *over a lifetime*
- ▶ What relationship do we expect for an annual population (life span = census interval)?
 - ▶ * $\mathcal{R} = \lambda$; each organism observed reproduces $\mathcal{R}$ offspring on average, all in one time step
- ▶ For a longer-lived population?
 - ▶ * The $\mathcal{R}$ offspring are produced slowly, so population changes slowly
 - ▶ * $\bar{\lambda}$ should be closer to 1 than $\mathcal{R}$ is.
 - ▶ * But on the same side (same answer about whether population is growing)

Studying population growth

- ▶ $\bar{\lambda}$ and $\mathcal{R}$ give related information about your population
- ▶ $\mathcal{R}$ is easier to calculate, and more generally useful
- ▶ But $\bar{\lambda}$ gives the actual rate of growth
 - ▶ More useful in cases where we expect the life table to be constant with exponential growth or decline for a long time

Growth and decline

- ▶ If we think a particular period of growth or decline is important, we might want to study how factors affect $\bar{\lambda}$
 - ▶ Complicated, but well-developed, theory
 - ▶ In a growing population, what happens early in life is more important to $\bar{\lambda}$ than to $\mathcal{R}$.
 - ▶ In a declining population, what happens late in life is more important to $\bar{\lambda}$ than to $\mathcal{R}$.
- ▶ Which phase is likely to be more important to ecology and evolution?
 - ▶ * The two phases (growth and decline) will be roughly balanced
 - ▶ * Because otherwise the population would go to zero or infinity

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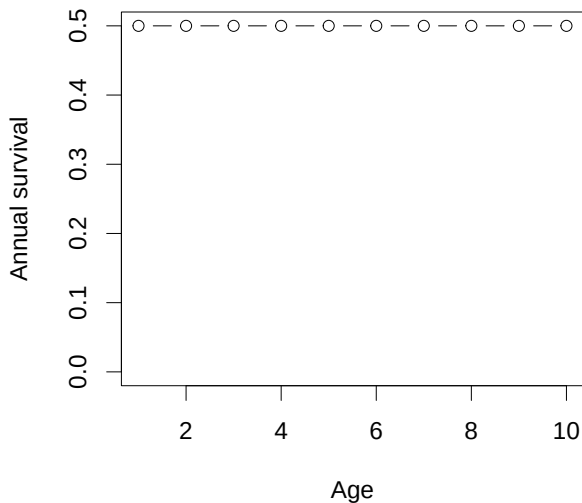
Patterns of survivorship

- ▶ What sort of patterns do you expect to see in p_x ?
 - ▶ * Younger individuals usually have lower survivorship
 - ▶ * Older individuals often have lower survivorship
- ▶ What about ℓ_x ?
 - ▶ * It goes down
 - ▶ * But sometimes faster and sometimes slower
 - ▶ * Best understood on a log scale

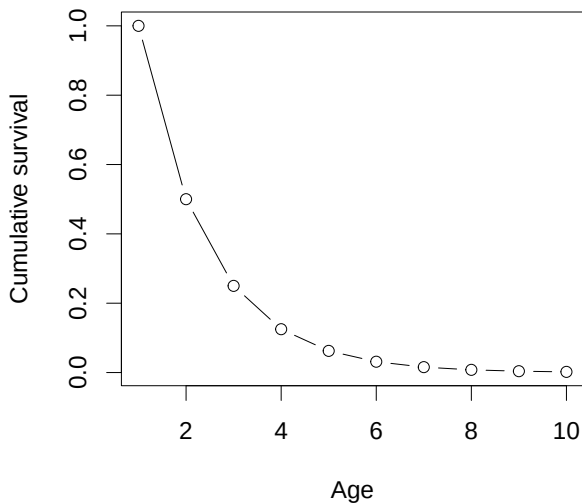
Starting off

- ▶ Recall: we always start from age *class* 1
 - ▶ If we count newborns, we still call them class 1.
- ▶ What is ℓ_1 when we count before reproduction?
 - ▶ * 1
 - ▶ * ℓ_1 is the probability you're counted at age class 1, *given that* you're counted at age class 1.
 - ▶ * We don't count individuals that we don't count

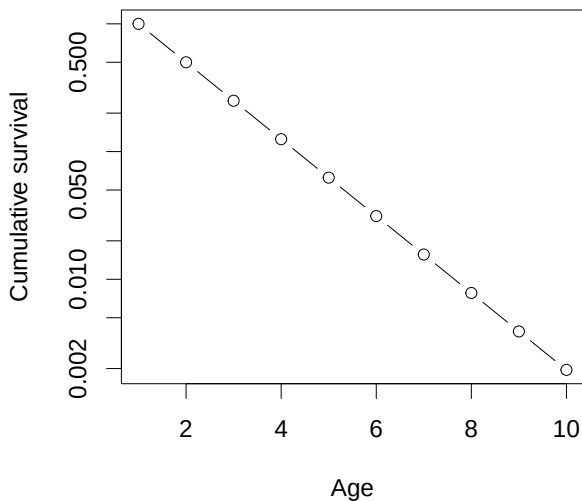
Constant survivorship (present)



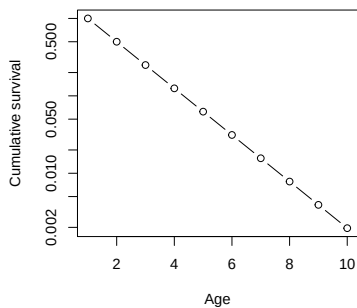
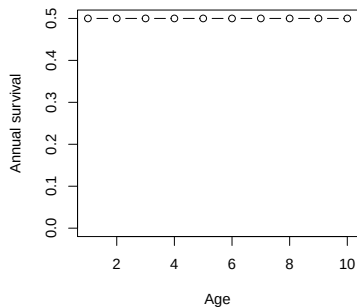
Constant survivorship (present)



Constant survivorship (present)



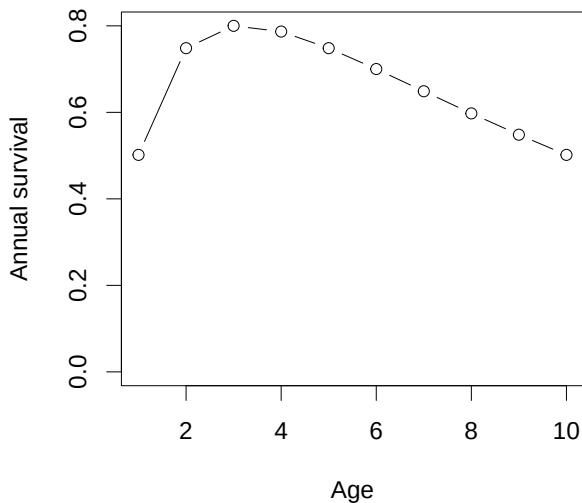
Constant survivorship



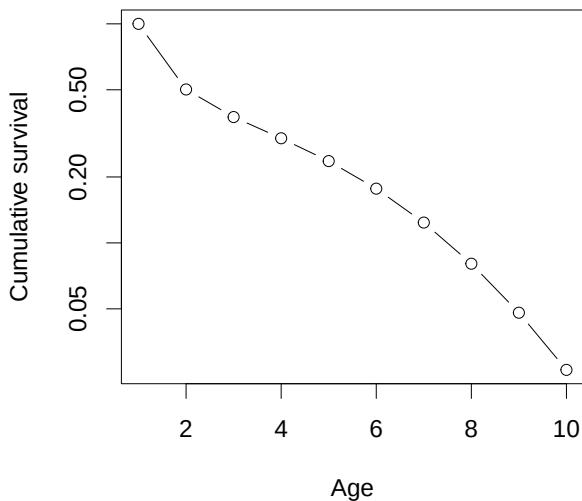
“Types” of survivorship

- ▶ There is a history of defining survivorship as:
 - ▶ Type I, II or III depending on whether it increases, stays constant or decreases with age (*don't memorize this, just be aware in case you encounter it later in life*).
 - ▶ Real populations tend to be more complicated
- ▶ Most common pattern is: high mortality at high and low ages, with less mortality between

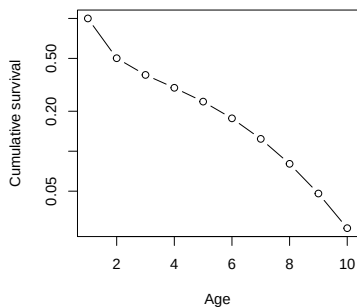
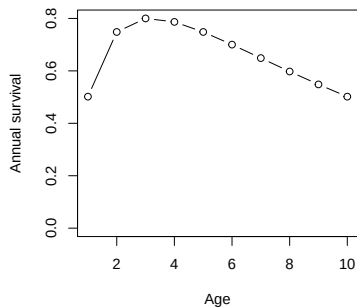
Changing survivorship (present)



Changing survivorship (present)



Changing survivorship



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Fecundity

- ▶ Just as in our simple population growth models, we define fecundity as the expected number of offspring at the next census produced by an individual observed at this census
 - ▶ Parent must survive from counting to reproduction
 - ▶ Parent must give birth
 - ▶ Offspring must survive from birth to counting
- ▶ Remember to think clearly about sex when necessary
 - ▶ Are we tracking females, or everyone?

Fecundity patterns

- ▶ f_x is the average number of new individuals *counted* next census per individual in age class x *counted* this census
- ▶ Fecundity often goes up early in life and then remains constant
 - ▶ * Most birds, many large mammals
- ▶ It may also go up and then come down
 - ▶ * people
- ▶ It may also go up and up as organisms get older
 - ▶ * Many fish, many trees

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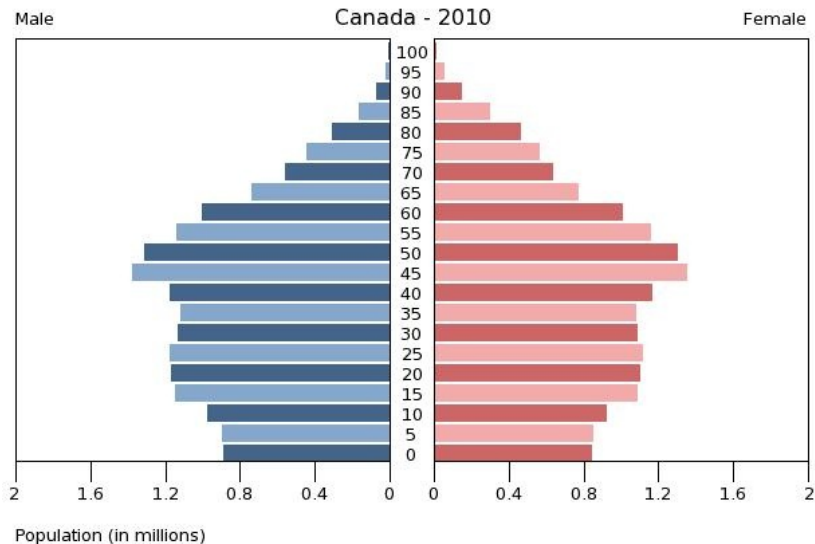
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CSLIDE Age distributions



Learning from the model

- ▶ If a population has constant size (ie., the same number of individuals are born every year), what determines the proportion of individuals in each age class?
 - ▶ * Distribution should be proportional to ℓ_x
- ▶ What if population is growing?
 - ▶ * We expect proportionally more individuals in younger age classes
 - ▶ * Number of births in more recent cohorts is larger

Stable age distribution

- ▶ If a population has reached a SAD, and is increasing at rate λ (given by the Euler equation):
 - ▶ the x year old cohort, born x years ago originally had a size λ^{-x} relative to the current one
 - ▶ a proportion ℓ_x of this cohort has survived
 - ▶ thus, the relative size of cohort x is $\lambda^{-x}\ell_x$
 - ▶ SAD depends only on survival distribution ℓ_x and λ .

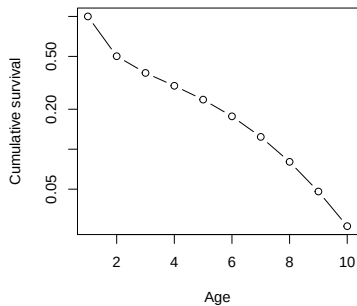
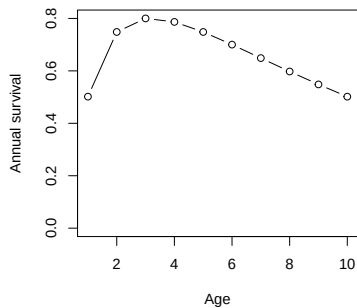
Patterns

- ▶ Populations tend to be bottom-heavy (more individuals at lower age classes)
 - ▶ Many individuals born, few survive to older age classes
- ▶ If population is growing, this increases the lower classes further
 - ▶ More individuals born more recently
- ▶ If population is *declining*, this shifts the age distribution in the opposite direction
 - ▶ Results can be complicated
 - ▶ Declining populations may be bottom-heavy, top-heavy or just jumbled

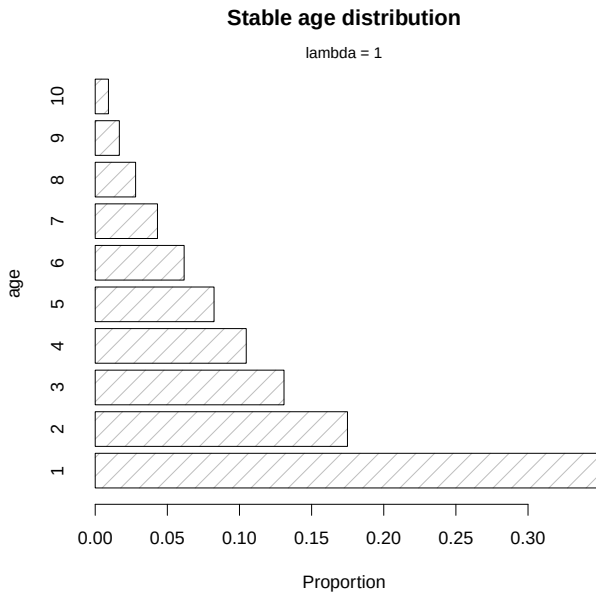
University cohorts

- ▶ McMaster accepts only first-year students. For any given stage (e.g., end of third year) the same proportion drop out each year
- ▶ What can you say about the relative size of the classes if:
 - ▶ The same number of students is admitted each year
 - ▶ * The lower classes are larger
 - ▶ More students are admitted each year
 - ▶ * The lower classes are larger (even more so)
 - ▶ Fewer students are admitted each year
 - ▶ * Anything could happen (drop outs and size change are operating in different directions)

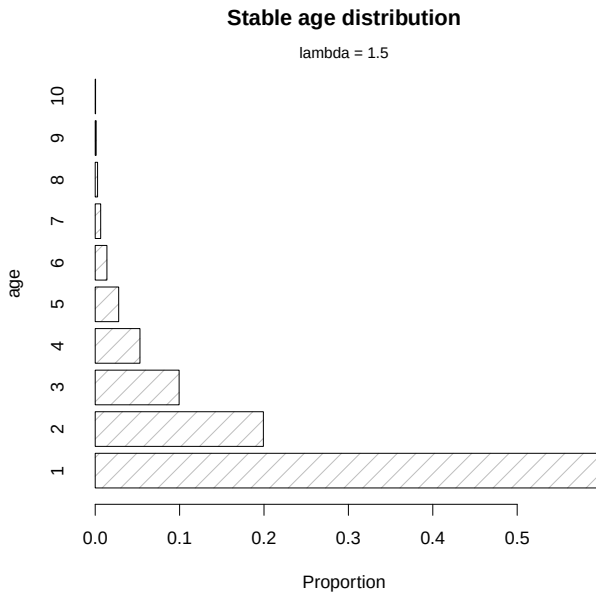
REPSLIDE Changing survivorship



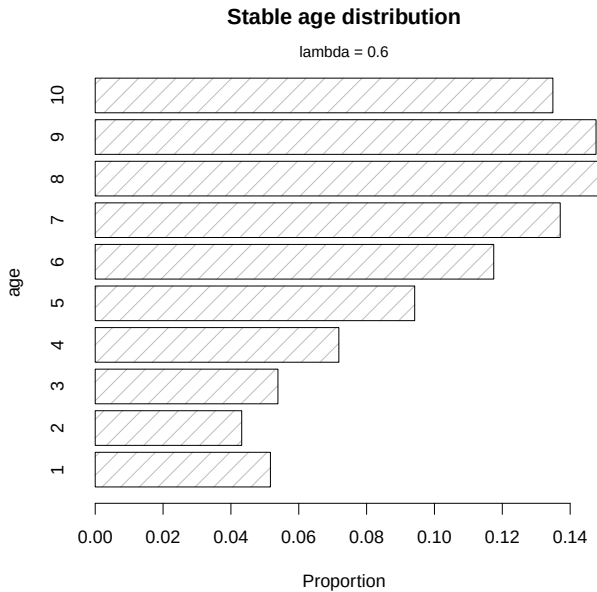
Age distributions



Age distributions



Age distributions



REPSLIDE The Euler equation

- ▶ If the life table doesn't change, then $\bar{\lambda}$ is given by
$$\sum_x \ell_x f_x \bar{\lambda}^{-x} = 1$$
- ▶ If the population has the structure we would expect from growing at rate λ , would it continue to grow at rate λ ?
- ▶ On the left-side each cohort started as λ times smaller than the one after it
 - ▶ Then got multiplied by ℓ_x .
- ▶ Under this assumption, is the next generation λ times bigger again?

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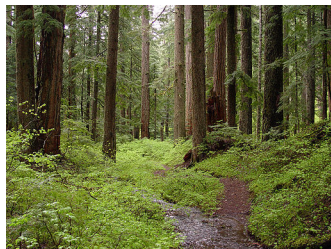
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Forest example

- ▶ Forests have obvious population structure
- ▶ They also seem to remain stable for long periods of time
- ▶ Populations are presumably *regulated* at some time scale



Forest size classes

- ▶ When we go to an apparently stable forest ecosystem, it seems to be dominated by large trees, not small ones. What up?
- ▶ How is it possible that these systems are dominated by large trees?
 - ▶ * Large trees are larger
 - ▶ * Population may be declining
 - ▶ * Trees may spend longer in some size classes than in others
 - ▶ * Life table may not be constant (smaller trees may recruit at certain times and places)

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Stage structure

- ▶ Stage structure works just like age structure, except that what stage you are in is not strictly predicted by how old you are
 - ▶ Age-structured models need fecundity, and survival probability
 - ▶ * In stage-structured models survival is typically broken into:
 - ▶ * Survival into same stage
 - ▶ * Survival with recruitment (ie., to the next larger class of individuals)
 - ▶ More complicated models are also possible

Advantages

- ▶ Stage structured models don't need a maximum age
- ▶ Nor one box for every single age class

Unregulated growth

- ▶ What happens if you have a constant stage table (no regulation)?
 - ▶ Fecundity, and survival and recruitment probabilities are constant
- ▶ Similar to constant life table
 - ▶ Can calculate $\mathcal{R}$ and $\bar{\lambda}$ (will be consistent with each other)
 - ▶ Can calculate a stable stage distribution
 - ▶ *$\mathcal{R}$ is about the same as in age structured model*
- ▶ Unregulated growth cannot persist

Summary

- ▶ If the life table remains constant (no regulation or stochasticity):
 - ▶ Reach a stable age (or stage) distribution
 - ▶ Grow or decline with a constant λ
 - ▶ Factors behind age distribution can be understood

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Regulated growth

- ▶ Our models in this unit assume that individuals are independent
- ▶ In this case, we expect populations to grow (or decline) exponentially
- ▶ We do not expect that the long-term average value of $\mathcal{R}$ or $\bar{\lambda}$ will be exactly 1.
 - ▶ * So we need regulation in real life

The value of simple models

- ▶ There is a lot of mathematical study of unregulated, age-structured populations, but it should be taken with a grain of salt
 - ▶ * We know that real populations are regulated
 - ▶ * Populations can't increase or decrease exponentially for very long
- ▶ Understanding this behaviour is helpful:
 - ▶ interpreting age structures in real populations
 - ▶ beginning to understand more complicated systems

Regulation and structure

- ▶ We expect real populations to be regulated
- ▶ Under regulation, the value of $\bar{\lambda}$ under regulation *could* be exactly 1
- ▶ There is also likely to be substantial variation from year to year, due to changing conditions and other random-seeming forces

Dynamics

- ▶ We expect to see smooth behaviour in many cases
- ▶ Cycles and complex behaviour should arise for reasons similar to our unstructured models:
 - ▶ Delays in the system
 - ▶ Strong population response to density
- ▶ Age distribution will be determined by:
 - ▶ ℓ_x , and
 - ▶ whether the population has been growing or declining recently
 - ▶ *Not really covered this year*